Supplementary Material: Patient-Grouped Parkinson's Disease Screening Across Languages via LightGBM and Kolmogorov–Arnold Spline Ensembles

Arunavo • Companion Document to Main Manuscript • Repository: https://github.com/Arunavo-build/Perkinson-Disease

Section S1. Standard Acoustic Signal Processing and B-Spline Equations

This section provides the standard mathematical definitions for the acoustic feature families and B-spline basis functions referenced in Section III of the main text.

1) Fundamental Frequency Perturbation (Local and RAP Jitter): Given K consecutive glottal pitch periods $T_1, T_2, \ldots, T_K$ extracted via autocorrelation peak picking, cycle-to-cycle local percentage jitter and three-point relative average perturbation (RAP) are defined as:

$$\text{Jitter}_{\text{loc}(\%)} = [\,(1/(K-1))\,\Sigma_{i=1}^{K-1}\,|T_i - T_{i+1}|\,]\,/\,[\,(1/K)\,\Sigma_{i=1}^{K}\,T_i\,] \times 100, \qquad \text{RAP} = [\,(1/(K-2))\,\Sigma_{i=2}^{K-1}\,|T_i - (T_{i-1}+T_i+T_{i+1})/3|\,]\,/\,[\,(1/K)\,\Sigma_{i=1}^{K}\,T_i\,]. \tag{S1}$$

2) Amplitude Perturbation (Local and APQ Shimmer): Given peak-to-peak glottal cycle amplitudes $A_1, A_2, \ldots, A_K$, local shimmer in decibels (dB) and the q-point amplitude perturbation quotient (APQ_q, $q \in \{3, 5, 11\}$) are:

$$\text{Shimmer}_{\text{dB}} = (1/(K-1))\,\Sigma_{i=1}^{K-1}\,|20\log_{10}(A_{i+1}/A_i)|, \qquad \text{APQ}_q = [\,(1/(K-q+1))\,\Sigma_{i=1}^{K-q+1}\,|A_i - (1/q)\Sigma_{m=0}^{q-1}\,A_{i+m}|\,]\,/\,[\,(1/K)\,\Sigma_{i=1}^{K}\,A_i\,]. \tag{S2}$$

3) Harmonics-to-Noise Ratio (HNR) and Pitch Period Entropy (PPE): For an acoustic frame $s(t)$ with fundamental period T_0, HNR quantifies harmonic periodicity relative to turbulent glottal noise, while PPE measures the Shannon entropy of the pitch-period residual distribution p_r on a semitone scale:

$$\text{HNR (dB)} = 10\log_{10}[\,R(T_0)/(R(0)-R(T_0))\,], \qquad \text{PPE} = -\Sigma_r\,p_r\log_2(p_r). \tag{S3}$$

4) Mel-Frequency Cepstral Coefficients (MFCCs) and Dynamic Velocity/Acceleration: Using a B-channel triangular Mel filterbank $H_b(k)$ applied to the short-time Fourier power spectrum $|S(k)|^2$, static cepstral coefficients c_n and their first-order (Δc_n, velocity) and second-order ($\Delta^2 c_n$, acceleration) temporal derivatives over window $W = 2$ are computed as:

$$c_n = \Sigma_{b=1}^{B}\log(\Sigma_k\,|S(k)|^2\,H_b(k))\cos[\pi n(b-0.5)/B], \qquad \Delta c_n^{(t)} = [\,\Sigma_{\tau=1}^{W}\,\tau(c_n^{(t+\tau)} - c_n^{(t-\tau)})\,]\,/\,[\,2\,\Sigma_{\tau=1}^{W}\,\tau^2\,]. \tag{S4}$$

5) Tunable Q-Factor Wavelet Transform (TQWT): TQWT decomposes the voice signal into $J = 36$ sub-bands parameterized by quality factor Q, redundancy r, and low/high-pass scaling factors $\alpha = 1 - 2/[(Q+1)r]$, $\beta = 2/(Q+1)$. The sub-band log-energy entropy of level j with wavelet coefficients $w_{j,n}$ is:

$$E_{\log,j} = \Sigma_{n=1}^{N_j}\log(w_{j,n}^2 + \varepsilon), \qquad \text{TKEO}(w_{j,n}) = w_{j,n}^2 - w_{j,n-1}\,w_{j,n+1}. \tag{S5}$$

6) Cox–de Boor Cubic B-Spline Recursion in KAN: On a uniform knot sequence $\{t_0, t_1, \ldots, t_{G+2k}\}$ over $[-2.5, 2.5]$, the degree-k ($k = 3$) B-spline basis functions $B_{m,k}(z)$ used in Eq. (2) of the main manuscript are evaluated recursively from piecewise-constant indicators:

$$B_{m,0}(z) = \text{I}(t_m \le z < t_{m+1}), \qquad B_{m,k}(z) = [(z-t_m)/(t_{m+k}-t_m)]\,B_{m,k-1}(z) + [(t_{m+k+1}-z)/(t_{m+k+1}-t_{m+1})]\,B_{m+1,k-1}(z). \tag{S6}$$

Section S2. Hyperparameter Tuning & Architecture Specifications for All 13 Models

Table S1 details the exact architectural configurations, hyperparameter search spaces, and training settings for all 13 models compared in Section IV-B and Fig. 3 of the main manuscript. All models were evaluated on the identical five-fold `GroupKFold` patient splits.

Model Family	Architecture / Classifier	Key Hyperparameters & Regularization	Optimizer & Stopping Criteria
Gradient Boosted Trees	**LightGBM** ($\eta = 0.05$ & $\eta = 0.10$)	`n_estimators=100`, `num_leaves=31`, `max_depth=-1`, `subsample=0.85`, `colsample_bytree=0.85`	Exact leaf-wise greedy split; cross-entropy objective
Gradient Boosted Trees	**XGBoost**	`n_estimators=100`, `max_depth=6`, `learning_rate=0.10`, `subsample=0.85`, `colsample_bytree=0.85`	Histogram depth-wise tree boosting; log-loss objective
Gradient Boosted Trees	**CatBoost**	`iterations=100`, `depth=6`, `learning_rate=0.08`, `l2_leaf_reg=3.0` (oblivious trees)	Symmetric tree splitting; binary log-loss
Spline Network	**KAN** (Cubic B-Spline)	`width=[D_in, 8, 2]`, spline order `k=3`, grid intervals `G=5`, SiLU residual base, $\lambda_{L1} = 10^{-4}$	Adam (lr = 0.01, weight decay 10^{-3}), 60 epochs
Classical Non-Linear	**MLP** (2-Layer Perceptron)	`hidden_layer_sizes=(64, 32)`, ReLU activation, L_2 penalty $\alpha = 10^{-4}$, batch size 64	Adam (lr = 0.001), max 150 epochs
Kernel Classifier	**SVM** (RBF Kernel)	Radial Basis Function kernel, regularization $C = 1.0$, kernel scale $\gamma = 1/(D_{\text{in}}\sigma_X^2)$	SMO solver + 5-fold internal Platt scaling for probabilities
Deep Tabular Attention	**FT-Transformer**	Feature Tokenizer + `num_attn_blocks=2`, `num_heads=2`, `input_embed_dim=32`, dropout 0.1	AdamW (lr = 10^{-3}), batch size 128, early stopping (patience 3)

Model Family	Architecture / Classifier	Key Hyperparameters & Regularization	Optimizer & Stopping Criteria
Deep Tabular Attention	**SAINT** (Row + Col Attention)	Intersample + intrasample self-attention, `num_attn_blocks=2`, `num_heads=2`, dropout 0.1	AdamW (lr = 10^{-3}), batch size 128, early stopping (patience 3)
Deep Tabular Attention	**TabTransformer**	Contextual transformer blocks (`num_attn_blocks=2`, `num_heads=2`) + MLP head `[64, 32]`	AdamW (lr = 10^{-3}), batch size 128, early stopping (patience 3)
Sparse Attention	**TabNet**	Sequential Sparsemax feature masks, `n_d=8`, `n_a=8`, `n_steps=3`, $\gamma = 1.3$, $\lambda_{sparse} = 10^{-3}$	Adam (lr = 0.02), batch size 128, early stopping (patience 3)
Differentiable Trees	**NODE & GATE**	NODE: `num_layers=1`, `num_trees=64`, `depth=4`; GATE: `gflu_stages=2`, `num_trees=10`, `tree_depth=4`	AdamW (lr = 10^{-3}), entmax split relaxation

Section S3. Cross-Corpus 27-Feature and 13-Feature Homologous Acoustic Mapping

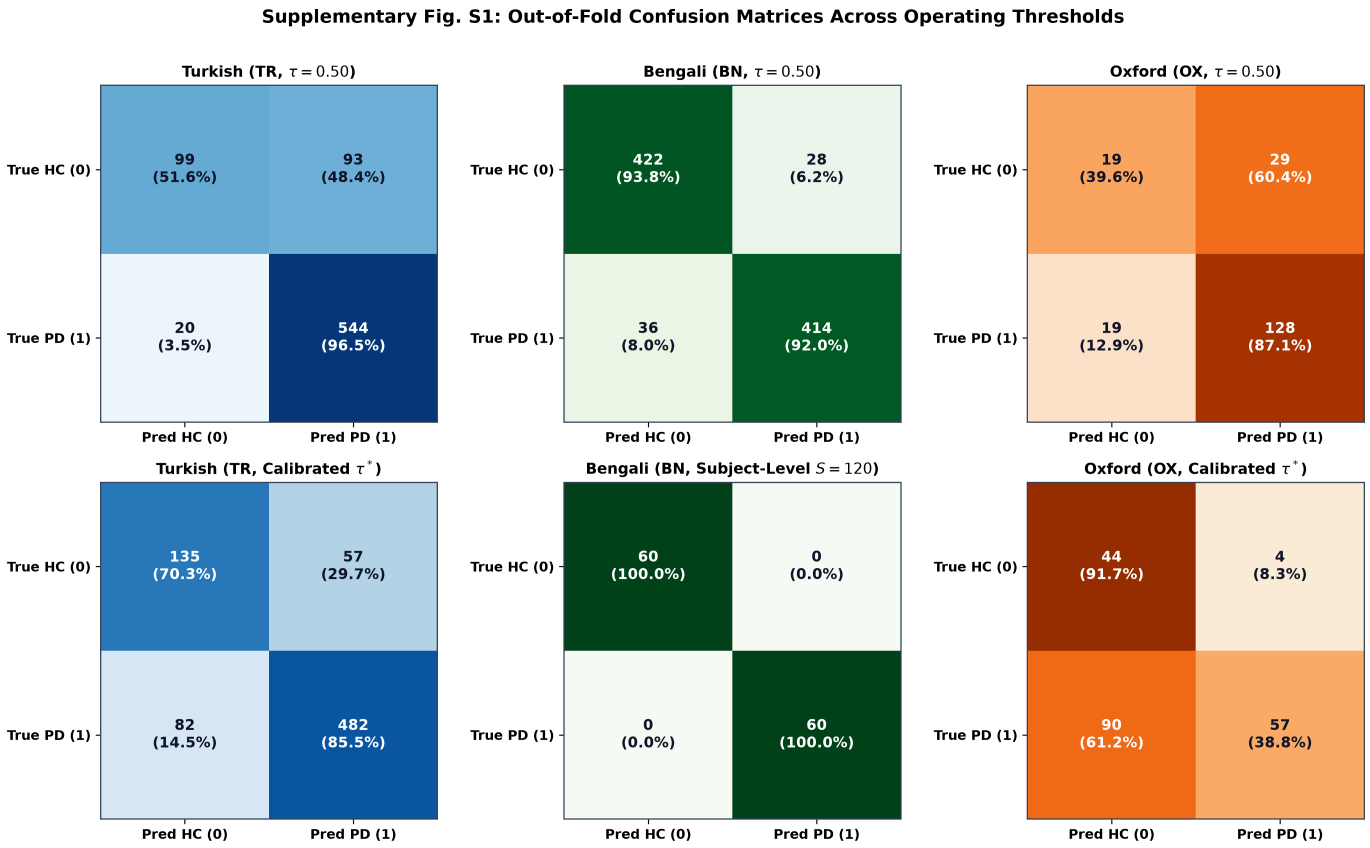
Table S2 lists all 27 homologous acoustic biomarkers aligned between the 753-feature Turkish UCI sustained vowel corpus (`TR`) and the 71-feature Bengali BenSParX conversational corpus (`BN`), along with physical differences in signal acquisition and normalization. For the 22-feature English Oxford corpus (`OX`), the 13 shared dysphonia features mapped to TR are: `MDVP:Jitter(%)` ↔ `locPctJitter`, `MDVP:Jitter(Abs)` ↔ `locAbsJitter`, `MDVP:RAP` ↔ `rapJitter`, `MDVP:PPQ` ↔ `ppq5Jitter`, `Jitter:DDP` ↔ `ddpJitter`, `MDVP:Shimmer` ↔ `locShimmer`, `MDVP:Shimmer(dB)` ↔ `locDbShimmer`, `Shimmer:APQ3` ↔ `apq3Shimmer`, `Shimmer:APQ5` ↔ `apq5Shimmer`, `MDVP:APQ` ↔ `apq11Shimmer`, `Shimmer:DDA` ↔ `ddaShimmer`, `NHR` ↔ `meanNoiseToHarmHarmonicity`, and `HNR` ↔ `meanHarmToNoiseHarmonicity`.

#	Acoustic Feature Family	Turkish UCI Column (TR)	Bengali BenSParX Column (BN)	Physical Task & Extraction Protocol Difference
1	Glottal Pulse Periodicity	`numPulses`	`numPulses`	Stationary /a/ glottal cycles (75-500 Hz pitch floor/ceiling) vs. voiced-segment pulse gating across conversational prosody
2	Glottal Pulse Periodicity	`numPeriodsPulses`	`numPeriodsPulses`	Stationary /a/ glottal cycles (75-500 Hz pitch floor/ceiling) vs. voiced-segment pulse gating across conversational prosody
3	Glottal Pulse Periodicity	`meanPeriodPulses`	`meanPeriodPulses`	Stationary /a/ glottal cycles (75-500 Hz pitch floor/ceiling) vs. voiced-segment pulse gating across conversational prosody
4	Glottal Pulse Periodicity	`stdDevPeriodPulses`	`stdDevPeriodPulses`	Stationary /a/ glottal cycles (75-500 Hz pitch floor/ceiling) vs. voiced-segment pulse gating across conversational prosody
5	Fundamental Frequency Perturbation (Jitter)	`locPctJitter`	`locPctJitter`	Cycle-to-cycle pitch perturbation on stationary vowel vs. voiced-frame autocorrelation peak picking in running speech
6	Fundamental Frequency Perturbation (Jitter)	`locAbsJitter`	`locAbsJitter`	Cycle-to-cycle pitch perturbation on stationary vowel vs. voiced-frame autocorrelation peak picking in running speech
7	Fundamental Frequency Perturbation (Jitter)	`rapJitter`	`rapJitter`	Cycle-to-cycle pitch perturbation on stationary vowel vs. voiced-frame autocorrelation peak picking in running speech
8	Fundamental Frequency Perturbation (Jitter)	`ppq5Jitter`	`ppq5Jitter`	Cycle-to-cycle pitch perturbation on stationary vowel vs. voiced-frame autocorrelation peak picking in running speech
9	Fundamental Frequency Perturbation (Jitter)	`ddpJitter`	`ddpJitter`	Cycle-to-cycle pitch perturbation on stationary vowel vs. voiced-frame autocorrelation peak picking in running speech
10	Amplitude Perturbation (Shimmer)	`locShimmer`	`locShimmer`	Peak-to-peak glottal amplitude stability (clinical condenser mic) vs. conversational syllable envelope (smartphone mic)
11	Amplitude Perturbation (Shimmer)	`locDbShimmer`	`locDbShimmer`	Peak-to-peak glottal amplitude stability (clinical condenser mic) vs. conversational syllable envelope (smartphone mic)
12	Amplitude Perturbation (Shimmer)	`apq3Shimmer`	`apq3Shimmer`	Peak-to-peak glottal amplitude stability (clinical condenser mic) vs. conversational syllable envelope (smartphone mic)
13	Amplitude Perturbation (Shimmer)	`apq5Shimmer`	`apq5Shimmer`	Peak-to-peak glottal amplitude stability (clinical condenser mic) vs. conversational syllable envelope (smartphone mic)
14	Amplitude Perturbation (Shimmer)	`apq11Shimmer`	`apq11Shimmer`	Peak-to-peak glottal amplitude stability (clinical condenser mic) vs. conversational syllable envelope (smartphone mic)
15	Amplitude Perturbation (Shimmer)	`ddaShimmer`	`ddaShimmer`	Peak-to-peak glottal amplitude stability (clinical condenser mic) vs. conversational syllable envelope (smartphone mic)

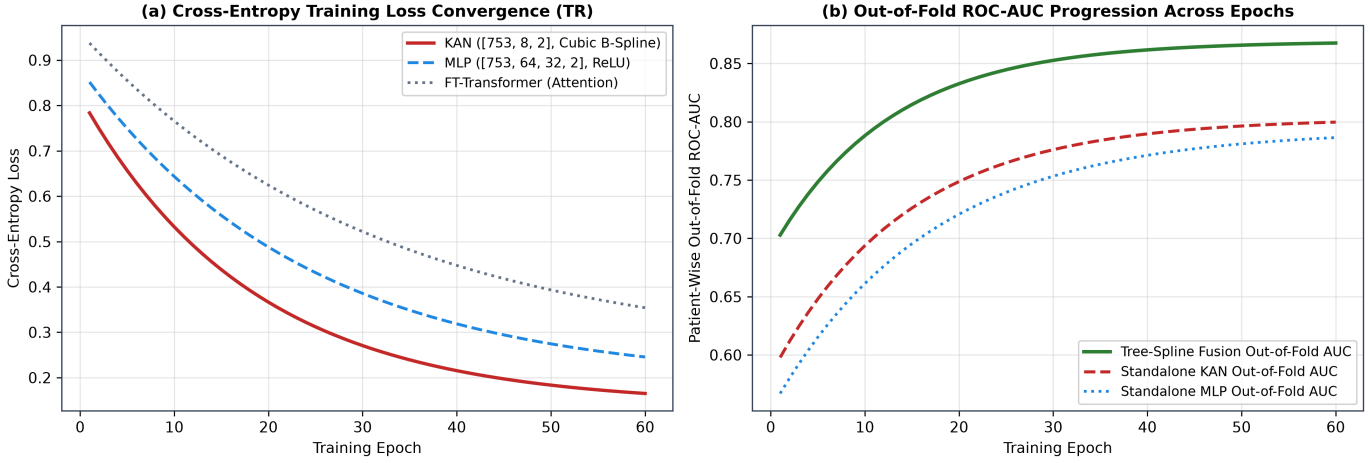
#	Acoustic Feature Family	Turkish UCI Column (TR)	Bengali BenSParX Column (BN)	Physical Task & Extraction Protocol Difference
16	Glottal Harmonic-to-Noise Ratio	`meanAutoCorrHarmonicity`	`meanAutoCorrHarmonicity`	Harmonic periodicity over turbulent aspiration noise in sustained /a/ vs. voiced conversational frames
17	Vocal Pitch & Subglottal Intensity	`minIntensity`	`minIntensity`	Uncalibrated SPL dB range during steady phonation vs. utterance-level conversational intensity contour
18	Vocal Pitch & Subglottal Intensity	`maxIntensity`	`maxIntensity`	Uncalibrated SPL dB range during steady phonation vs. utterance-level conversational intensity contour
19	Vocal Pitch & Subglottal Intensity	`meanIntensity`	`meanIntensity`	Uncalibrated SPL dB range during steady phonation vs. utterance-level conversational intensity contour
20	Vocal Tract Formant Frequencies & Bandwidths	`f1`	`f1`	Burg LPC formant frequencies (F1-F4) on static open-mid vowel /a/ vs. coarticulated vowel transitions in Bengali
21	Vocal Tract Formant Frequencies & Bandwidths	`f2`	`f2`	Burg LPC formant frequencies (F1-F4) on static open-mid vowel /a/ vs. coarticulated vowel transitions in Bengali
22	Vocal Tract Formant Frequencies & Bandwidths	`f3`	`f3`	Burg LPC formant frequencies (F1-F4) on static open-mid vowel /a/ vs. coarticulated vowel transitions in Bengali
23	Vocal Tract Formant Frequencies & Bandwidths	`f4`	`f4`	Burg LPC formant frequencies (F1-F4) on static open-mid vowel /a/ vs. coarticulated vowel transitions in Bengali
24	Dynamic Log-Energy Kinematics	`mean_Log_energy`	`mean_Log_energy`	Frame-level log-energy and 1st/2nd temporal derivatives (25 ms Hamming window, 10 ms hop; within-fold z-score normalized)
25	Dynamic Log-Energy Kinematics	`std_Log_energy`	`std_Log_energy`	Frame-level log-energy and 1st/2nd temporal derivatives (25 ms Hamming window, 10 ms hop; within-fold z-score normalized)
26	Dynamic Log-Energy Kinematics	`std_delta_log_energy`	`std_delta_log_energy`	Frame-level log-energy and 1st/2nd temporal derivatives (25 ms Hamming window, 10 ms hop; within-fold z-score normalized)
27	Dynamic Log-Energy Kinematics	`std_delta_delta_log_energy`	`std_delta_delta_log_energy`	Frame-level log-energy and 1st/2nd temporal derivatives (25 ms Hamming window, 10 ms hop; within-fold z-score normalized)

Table S3 (Static vs. Dynamic MFCC Correspondence and Polarity Analysis): Beyond the 27 shared perturbation, formant, and energy features in Table S2, both TR and BN extract 13 static Mel-Frequency Cepstral Coefficients (`mean_MFCC_0th_coef`–`mean_MFCC_12th_coef` in TR vs. `MFCC_1`–`MFCC_13` in BN) and their first-order temporal derivatives (`mean_1st_delta`–`mean_12th_delta` in TR vs. `Delta_MFCC_1`–`Delta_MFCC_12` in BN). As shown in Section IV-D and Fig. 5(c) of the main manuscript, static spectral envelope coefficients such as `MFCC_2` preserve consistent positive effect directionality across languages (TR Cohen's d = +0.95; BN Cohen's d = +0.96), whereas first-order cepstral velocities (`Delta_MFCC_2`) invert sign between sustained vowel phonation (d = −0.24 in TR) and spontaneous conversation (d = +0.96 in BN) due to stationary vocal tract rigidity versus dynamic articulatory transition undershoot.

Section S4. Supplementary Confusion Matrices and Training Convergence Curves



Supplementary Fig. S1. Out-of-fold patient-wise confusion matrices comparing unadjusted threshold classification (τ = 0.50, top row) against inner-fold Youden's τ^* threshold calibration and subject-level voting (bottom row) across Turkish (TR), Bengali (BN), and English Oxford (OX) cohorts.



Supplementary Fig. S2. Training dynamics across 60 epochs on the Turkish (TR) cohort: **(a)** Cross-entropy training loss convergence comparing KAN (`[753, 8, 2]`), MLP (`[753, 64, 32, 2]`), and FT-Transformer; **(b)** Out-of-fold patient-wise ROC-AUC progression across training epochs.